FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – III
PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A**	19.	**B**	37.	**D**
2.	**A**	20.	**A**	38.	**B**
3.	**B**	21.	**D**	39.	**C**
4.	**C**	22.	**A**	40.	**D**
5.	**B**	23.	**C**	41.	**B**
6.	**BC**	24.	**ABC**	42.	**BC**
7.	**AD**	25.	**BC**	43.	**ABCD**
8.	**BD**	26.	**ABCD**	44.	**ABC**
9.	**ABC**	27.	**BCD**	45.	**BC**
10.	**ABC**	28.	**AB**	46.	**AB**
11.	**ACD**	29.	**D**	47.	**ACD**
12.	**ABC**	30.	**AB**	48.	**ABD**
13.	**ABC**	31.	**B**	49.	**ABC**
14.	**6**	32.	**9**	50.	**1**
15.	**8**	33.	**8**	51.	**5**
16.	**4**	34.	**3**	52.	**9**
17.	**2**	35.	**5**	53.	**1**
18.	**6**	36.	**3**	54.	**4**

Physics

PART – I

SECTION – A

1. A

During 0 to t_0, potential across capacitor is zero i.e., charge on it is zero. Therefore there is no current through it. It means $V_{CD} = 0$. During t_0 to $2t_0$, voltage, therefore charge on capacitor is increasing linearly hence direct current is flowing through it therefore V_{CD} = constant.
During $2t_0$ to $3t_0$, V = constant $\Rightarrow i = 0$ and during $3t_0$ to $5t_0$ voltage decreases.

2. A

$$\left.\begin{aligned} S_A &= v_0 t - \frac{1}{2}\mu_A . g t^2 \\ S_B &= v_0 t - \frac{1}{2}\mu . g t^2 \end{aligned}\right] t = \sqrt{\frac{2(L-\ell)}{(\mu_B - \mu_A)g}}$$

$$S_A - S_B = L - \ell$$

3. B

$W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$

$$W_{AB} = \int_{V_A}^{V_B} P dV = \int_{V_A}^{V_B} \frac{C}{V^2} dV = -\frac{C}{V}\Big|_{V_A}^{V_B} = PV\Big|_A^B = P_A V_A - P_B V_B = -PV$$

$$W_{BC} = P_B\left(V_C - V_B\right) = 4P\left[\frac{V}{6} - \frac{V}{2}\right] = -\frac{4}{3}PV$$

$W_{CD} = 0$

$$W_{DA} = P_A(V_A - V_D) = P_A V_A - P_A V_D = PV - P\frac{V}{6} = \frac{5PV}{6}$$

$$\therefore W = -PV - \frac{4}{3}PV + \frac{5}{6}PV = -\frac{3}{2}PV$$

$\therefore$ Work done on the gas is $\frac{3}{2}PV$

4. C

Because $V_P = V_Q$

5 B

$$2\pi\sqrt{\frac{mb}{2k(b-\ell)}}$$

6. B, C

7. A, D

characteristic x-rays depend only on the atomic number z.

$\therefore \lambda_1 = \lambda_2 = \lambda_3$

8. B,D

$$(K.E)_{max} = \frac{hc}{\lambda} - \phi$$

9. A,B,C

10. A, B, C

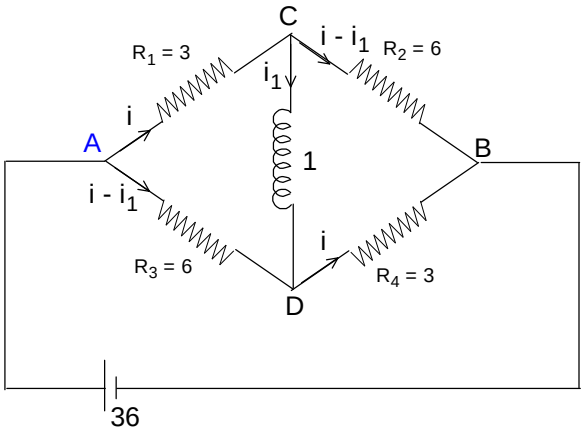
$f_{max} = f_0\left(\frac{v+v_D}{v-v_s}\right), f_{min} = f_0\left(\frac{v-v_D}{v+v_S}\right)$

$v_s = v_D = a\omega = a2\pi f = 1\times 2\pi \times 2 = 4\pi$

$f_{max} = 323Hz \;\; f_{min} = 278\ Hz$

$\Delta\phi$(between S and D) $= \pi$

11. (A), (C) & (D)



using Kirchhoff's loop law

$-3i - 1\frac{di}{dt} - 3i + 36 = 0$

$\therefore \frac{di}{dt} = 36 - 6i \ldots\ldots\ldots\ldots(i)$

$-3i - 6(i - i_1) + 36 = 0$

$\Rightarrow 3i - 2i_1 = 12 \ldots\ldots\ldots(ii)$

by (i) and (ii) $\frac{di_1}{dt} = 12 - 4i_1$

on solving

$i_1 = 3\left(1 - e^{-4t}\right)$

from (ii) $i = 2\left(1 - e^{-4t}\right) + 4$

$\therefore$ power supply by battery = $36(2i - i_1) = 36(9 - e^{-4t})$

$(i - i_1)_{t=\infty} = 3A$

12. ABC

$\left.\begin{array}{l} X_L = 2X_C \\ X_C' = 4X_C \\ X_L' = 2X_L \end{array}\right] X_L' - X_C' = 2X_L - 4X_C = 2[X_L - 2X_C] = 0$

13. ABC

The potential of the two surface will be equal when charge Q flows from inner to outer shell.

SECTION – C

14. 6

$^{14}_{4}N \longrightarrow ^{10}_{5}B + ^{4}_{2}He$

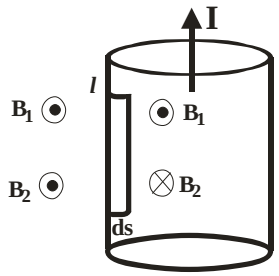
$2 = (0.00307 - 0.01294 - 0.00260)931$

$2 = 2.01247 \times 931 = 11.60$ MeV

15. 8

Consider a small rectangular strip on the inner cylinder.

Let B_1, is magnetic field due to rest of the cylinder excluding the strip and B_2 is magnetic field due to strip.



$B_2 - B_1 = 0$ (Inside)

$B_2 + B_1 = \frac{\mu_0}{2\pi}\frac{I}{a}$(outside)

$\Rightarrow \mathbf{B}_1 = B_2 = \frac{\mu_0}{4\pi}\frac{I}{a}$

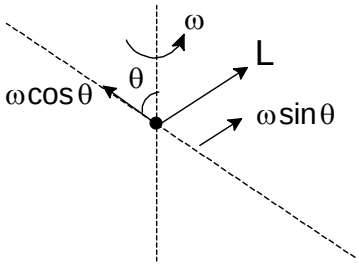
$\therefore \quad F = (Jds)\vec{l} \times \vec{B_1}$

$F = \left(\frac{I}{2\pi a}\right)ds l \frac{\mu_0}{4\pi}\frac{I}{a}$

$\therefore \quad P = \frac{F}{lds} = \frac{\mu_0 I^2}{8\pi^2 a^2}$

16. (4)

17. K = 2



Magnitude of angular momentum at any time is $L = \left(\frac{1}{12}m\ell^2\right).\omega\sin\theta.$

The tip of the vector L traces out a circle in a horizontal plane with frequency ω.

The radius of this circle is the horizontal component of L.

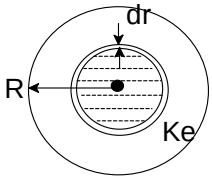
Which is $L_\perp = L\cos\theta$. The rate of change of L therefore has magnitude

$\left|\frac{dL}{dt}\right| = \omega L_\perp = \omega L\cos\theta = \omega\left(\frac{m\ell^2}{12}\omega\sin\theta\right)\cos\theta$ and it is directed into the page at the instant shown.

The torque due to the strings is $\tau = 2T\left(\frac{\ell}{2}\right)\cos\theta = \left|\frac{dL}{dt}\right|$

$T = \frac{m\ell}{12}\omega^2\sin\theta$

18. $\alpha = 6$



Thermal resistance of hollow sphere of inner radius r and outer radius R

$$R_{th} = \frac{(R-r)}{4\pi KRr}$$

If a shell of radius r and thickness dr freezes in time dt

Here

$$(dm)L_f = (q)dt$$

$$\delta_\omega \left(4\pi r^2 dr\right)L_f = \left(\frac{T}{\frac{(R-r)}{4\pi KRr}}\right)dt$$

$$\frac{\delta_\omega}{KT}\int_0^R r(R-r)dr = t$$

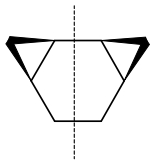
or $$t = \frac{\delta LR^2}{6KT}$$

Chemistry

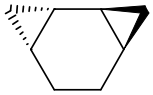
PART – II

SECTION – A

19. B



(1)



(2)

\+ enantiomer

(3)

20. A

$E_{photon} = 13.6 \times 4\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 10.2 \text{ eV}$

$10.2 = 13.6 \times 1\left(\frac{1}{12} - \frac{1}{n^2}\right) \Rightarrow n = 2$

$2\pi r = n\lambda$

$2\pi \times 0.529 \times \frac{n^2}{Z} = n\lambda$

$\lambda = 2\pi \times 0.529 \times \frac{2}{1} = 4\pi \times 0.529$

$\because (\Delta x)(\Delta\lambda) = \frac{\lambda^2}{4\pi}$

$\Delta\lambda = \frac{\lambda}{4\pi} = 0.529\,\overset{0}{A}$

21. D

$\left[IrBr_2(en)(H_2O)_2\right]^+ \Rightarrow G.I = 3, O.A = 2, O.I = 4$

$\left[CoBr_2(en)_2\right]^+ \Rightarrow G.I = 2, O.A = 2, O.I = 3$

$\left[PtBr(NO_2)(SCN)(NH_3)_3\right] \Rightarrow G.I = 4, O.A = 2, O.I = 5$

22. A

$CMC \propto \frac{1}{\text{size of tail of surfactant}}$

23. C

$FeS + H_2SO_4 \rightarrow FeSO_4 + \underset{(B)}{H_2S}\uparrow$

$2FeCl_3 + 3H_2S \longrightarrow 2FeCl_2 + 2HCl + \underset{(\text{White ppt.})}{S}\downarrow$

$(CH_3COO)_2Pb + H_2S \longrightarrow \underset{(\text{Black ppt.})}{PbS}\downarrow$

$MnO_4^- + H_2S + H^+ \longrightarrow Mn^{2+} + S\downarrow + H_2O$

$H_2S + Na_2\left[Fe(CN)_5NO\right] \longrightarrow$ purple colour

24. A, B, C

$Q_{12} = C_p(T_2 - T_1)$

$Q_{34} = C_p(T_4 - T_3)$

$Q_{56} = C_p(T_6 - T_5)$

$T_1 = T_3 = T_5 = T_7 = 298\ K$

For adiabatic step $2 \rightarrow 3$

$P_3^{1-\gamma} T_3^{\gamma} = P_2^{1-\gamma} T_2^{\gamma}$

$\left(\frac{32}{1}\right)^{\frac{1-\gamma}{\gamma}} T_3 = T_2$

$T_2 = (32)^{\frac{-2/3}{5/3}} T_3$

$= (32)^{\frac{-2}{5}} T_3 = \frac{1}{4} T_3 = \frac{1}{4} T_1$

$T_4 = (32)^{\frac{-2}{5}} T_5 = 2^{\frac{-2}{5}} T_3 = T_2 = T_6$

$Q_{total} = C_P \times 3(T_2 - T_1)$

$= \frac{5R}{2} \times 3\left(-\frac{3}{4} \times 300\right) = 45 \times 75 = -3375\,cal$

$\Delta S_{system} = 0 + 1 \times R\,In\frac{1}{(32)^3}$

$= -15\,R\,In\,2$

25. B, C

$Pb_3O_4 + 4HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

0.05 2 0 0 0 initial mols

0 1.8 0.1 0.05 final mol

$HNO_3 + NaOH \longrightarrow NaNO_3 + H_2O$

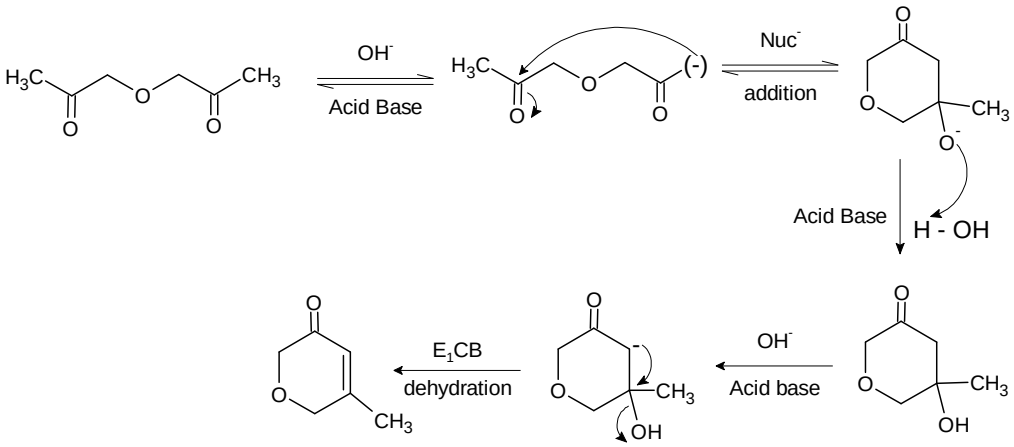
$1.8\,mol$

$\therefore$ mols of NaOH required $= 1.8$

Residue is PbO_2. Its mols formed $= 0.05$

$\therefore$ mass of $PbO_2 = 0.05 \times 239 = 11.95\,g$

26. A, B, C, D



27. B, C, D

$$H_2O_2 + 2-\text{ethylanthraquione} \underset{O_2}{\overset{H_2/Pd}{\rightleftharpoons}} 2-\text{ethylanthraquione}$$

28. A, B
Fact

29. D
In azeotrope two liquids are behaving as separate individuals.

30. A, B
(A) It is perkin reaction
(B) It is knoevenagel reaction
(C) It is based on reformatsky reaction

(D) $Ph-\underset{*}{\overset{OH}{\overset{|}{HC}}}-COOH$ will be product

31. B
Initial formula was $MgAl_2O_4$ and now formula is $MgAl_2X_3$, so anionic vacancy is 1.

SECTION – C

32. 9

$$\psi = \left[2/\left(\pi C^2\right)\right]^{3/4} e^{-\left(x^2+y^2+z^2\right)/C^2}$$

$$(\psi)^2\, dx\, dy\, dz = \left(2/\pi C^2\right)^{3/2} e^{-2}\left(x^2+y^2+z^2\right)/C^2\, dx\, dy\, dz$$

$$= \left[2/\left(4\pi nm^2\right)\right]^{3/2} e^{-2}\left[(1.2)^2+(-1)^2+(0)^2\right]/4\times(0.004\ nm)^3$$

$$= 1.2\times10^{-9} \qquad \therefore Y = 9$$

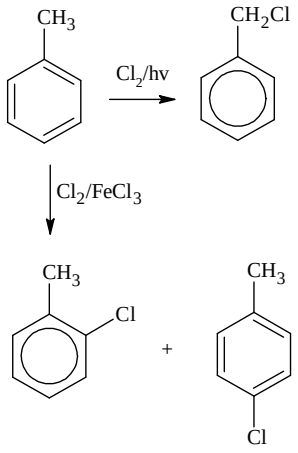
33. 8

$$NH_4NO_3(s) + \text{heat} \xrightarrow{H_2O} NH_4^+(aq) + NO_3^-(aq)$$

$$\text{Heat abosorbed} = ms\Delta T = 25\times1\times4 = 100\ \text{cal}$$

$$\Delta H_{solution} = \frac{100\times10^{-3}\,k\,cal}{12.5\times10^{-3}\,mol} = 8\ \text{kcal/mol}$$

34. 3
X is



35. 5

Ag_2SO_3, AgCl, $Ag_2C_2O_4$, $AgNO_2$ & CH3COOAg are white ppt., while Ag_3PO_4 is yellow and $Ag_2C_2O_4$ is red.

36. 3

$$\frac{a-P^2}{P^2} = b\ kt$$

$$\frac{a}{P^2} - 1 = b\ kt$$

on differentiating

$$-\frac{2a}{P^3}dP = b\ k\ dt$$

$$-\frac{dP}{dt} = \frac{bk}{2a}P^3$$

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. (D)

$\Delta EFM \sim \Delta AFD$

$\Rightarrow \frac{\text{area } \Delta EFM}{\text{area } \Delta AFD} = \left(\frac{MF}{FD}\right)^2$

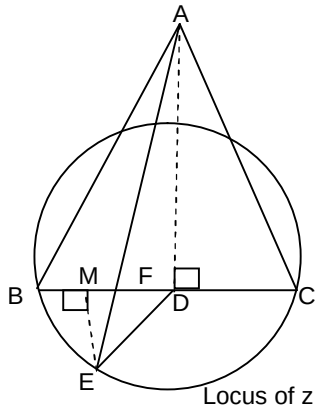
$\Rightarrow FD = 2MF$ ...(1)

$\Rightarrow BF = 2FD$

Let BD = a then area $\Delta EFD = \frac{1}{2} \times \frac{a}{3} \times a\frac{\sqrt{3}}{2} = \frac{a^2}{4\sqrt{3}}$

Area $\Delta AFC = \frac{1}{2} \times \left(\frac{4a}{3}\right) \times 2a.\frac{\sqrt{3}}{2} = \frac{2a^2}{\sqrt{3}}$

So $\frac{\text{area } \Delta FED}{\text{area } \Delta AFC} = \frac{1}{8}$ ....(2)



Now $|z - z_a|_{max} = a(1+\sqrt{3})$

$|z - z_a|_{min} = a\left(\sqrt{3}-1\right)$

$\Rightarrow \left(\frac{|z - z_a|_{max}}{|z - z_a|_{min}} - 2\right)^2 = 3$ .....(3)

from (2) & (3) k = 24.

38. (B)

$$S=\sum_{r=1}^{n}\left(\frac{\sin\theta}{2\cos(r+1)\theta\cos r\theta\sin\theta}\right)=\frac{1}{2\sin\theta}\sum_{r=1}^{n}\left(\frac{\sin(r+1)\theta-r\theta}{\cos(r+1)\theta\cos r\theta}\right)=\frac{1}{2\sin\theta}\left(\sum_{r=1}^{n}\left(\tan(r+1)\theta-\tan r\theta\right)\right)$$

$$=\frac{1}{2\sin\theta}\left(\left(\tan(n+1)\theta-\tan\theta\right)=\left(\frac{\sin n\theta}{\cos\left(n+1\right)\theta.\sin n\theta}\right)\right.$$

39. (C)

Any point on the line through $P(a,2)$ is $(a+r\cos\theta, 2+r\sin\theta)$ and meets the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at two distinct points A and D

$$\Rightarrow \frac{(a+r\cos\theta)^2}{9}+\frac{(2+r\sin\theta)^2}{4}=1$$

$$\Rightarrow (4\cos^2\theta+9\sin^2\theta)r^2+4(2a\cos\theta+9\sin\theta)r+4a^2=0$$

$$\Rightarrow PA.PD=\frac{4a^2}{(4\cos^2\theta+9\sin^2\theta)}$$

also line meets the axes at B and C $\Rightarrow$ PB.PC= $\dfrac{2a}{\sin\theta.\cos\theta}$

since PA, PB, PC and PD are in G.P.

$$\Rightarrow \frac{4a^2}{4\cos^2\theta+9\sin^2\theta}=\frac{2a}{\sin\theta.\cos\theta}$$

$$\Rightarrow 2a\sin2\theta+5\cos2\theta=13$$

$$\Rightarrow -1\le\frac{13}{\sqrt{4a^2+25}}\le1\Rightarrow a\le-6 \quad \text{or} \quad a\ge6$$

40. (D)

$$\text{As } \sum_{k=1}^{n}[(k+3)^2-(k+2)](k+2)! + 3.3! = 2003.2003!$$

$$\Rightarrow \sum_{k=1}^{n}[(k+3)(k+3)!-(k+2)(k+2)!] + 3.3! = 2003.2003!$$

$$\Rightarrow (n+3)(n+3)! = 2003.2003! \Rightarrow n = 2000$$

41. (B)

$$\int_0^{1/2} e^x\left[\left(\sin^{-1}x-\frac{1}{\sqrt{1-x^2}}\right)+\left(\frac{1}{\sqrt{1-x^2}}-\frac{x}{(1-x^2)^{3/2}}\right)\right]dx$$

$$=e^x\left(\sin^{-1}x-\frac{1}{\sqrt{1-x^2}}\right)\Big|_0^{1/2}$$

$$=e^{1/2}\left(\frac{\pi}{6}-\frac{2}{\sqrt{3}}\right)-1(0-1)$$

42. (B), (C)

Applying Leibnitz rule, we get

$$f(x) = x - x^2 f(x)$$

$$f(x)=\frac{x}{1+x^2}$$

$$\int_{-\frac{1}{2}}^{1}\frac{x}{1+x^2}dx=\int_{-\frac{1}{2}}^{1}\frac{x}{1+x^2}dx$$

$$=\frac{1}{2}\ln\frac{8}{5}$$

43. (A),(B), (C), (D)

$$\tan\frac{B}{2}=\frac{2}{3},\tan\frac{C}{2}=\frac{1}{2}\Rightarrow\tan\frac{A}{2}=\frac{4}{7}$$

$\tan\frac{B}{2}.\tan\frac{C}{2}=\frac{s-a}{s}\Rightarrow 2s=3a=42$

$\therefore$ Perimeter=42

$\therefore \Delta = r.s. = 84cm^2$

$\therefore \tan\frac{A}{2},\tan\frac{B}{2},\tan\frac{C}{2}$ all are less than 1. All angles are acute.

44. (A), (B), (C)

(B) $(a + c - b)1^2 + 2c\ (-1) + (b + c - a) = 0$

-1 is root of the equation which is rational. 2nd must also be rational

(C) $x=\frac{2}{c}$ Satisfies given equation which is rational

45. (B), (C)

$I=\frac{2}{1-e}\int\frac{dt}{t^2+\left(\frac{1+e}{1-e}\right)}$ (t = tanx/2)

if $0 < e < 1$, $\frac{1+e}{1-e}>0$, So, (B) is correct

If $e > 1$, $\frac{1+e}{1-e}<0$ So, (C) is correct.

46. (A), (B),

$$\frac{x-1}{\frac{0}{5}}=\frac{y-1}{\frac{4}{5}}=\frac{z+2}{\frac{-3}{5}}=\pm 1$$

47. (A), (C) ,(D)

$Y-y=\frac{dy}{dx}(X-x)$

$A\left(x-\frac{\frac{y}{dy}}{\frac{dx}{dx}},\ 0\right),\ B(x,\ 0)$

O B A

$x-\frac{\frac{y}{dy}}{\frac{dx}{dx}}=2x,\ -y=x\frac{dy}{dx}$

$\frac{dy}{y}=-\frac{dx}{x}$

LN |Y| = $-$LN|X| + C
CURVE IS PASSING FROM (1, 1), C = 0
XY = 1

48. (A), (B), (D)

Multiply R_1, R_2, R_3 respectively by a, b, c and take out a, b, c common from C_1, C_2, C_3 respectively.

$$\therefore \quad D = \begin{vmatrix} a^2+1 & a^2 & a^2 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix}$$

Apply $R_1 + R_2 + R_3$ and take $1+\Sigma a^2$ common

$$\therefore D = \left(1+\Sigma a^2\right)\begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix}$$

Make two zeros by $C_2 - C_1,\ C_3 - C_1$

$\therefore D = a^2+b^2+c^2+1 = 1(\text{given})$

$\Rightarrow a^2+b^2+c^2 = 0 \ \Rightarrow (c)$

49. (A), (B), (C),(D)

GIVEN $\frac{1}{2}\left(aP_1+bP_2+cP_3\right)=K \Rightarrow y=\frac{a}{P_1}+\frac{b}{P_2}+\frac{c}{P_3}$ IS MINIMUM.

WHEN $y=\frac{1}{2K}\left(aP_1+bP_2+cP_3\right)\left(\frac{a}{P_1}+\frac{b}{P_2}+\frac{c}{P_3}\right)$ IS MINIMUM.

BUT , $y=\frac{1}{2K}\left(a^2+b^2+c^2+ab\left(\frac{P_1}{P_2}+\frac{P_2}{P_1}\right)+bc\left(\frac{P_2}{P_3}+\frac{P_3}{P_2}\right)+ab\left(\frac{P_1}{P_3}+\frac{P_3}{P_1}\right)\right)$

$\geq \frac{1}{2K}\left(a^2+b^2+c^2+2ab+2bc+2ac\right)$

$\Rightarrow y \geq \frac{(a+b+c)^2}{2K}$ when $\frac{P_1}{P_2}=\frac{P_2}{P_1}=\frac{P_2}{P_3}=\frac{P_3}{P_2}=\frac{P_1}{P_3}=\frac{P_3}{P_1}=1$

$\Rightarrow$ i.e, when $P_1=P_2=P_3$

$\therefore$ P is incentre of ΔABC

SECTION – C

50. (1)

Let AD, BE & CF are medians M_1, M_2 & M_3 respectively

Clearly $M_1 = AD = \frac{1}{2}\sqrt{2(b^2+c^2)-a^2}$ $\quad M_2 = \frac{1}{2}\sqrt{2(a^2+c^2)-b^2}$ $\quad M_3 = \frac{1}{2}\sqrt{2(a^2+b^2)-c^2}$

$\Rightarrow M_1^2 + M_2^2 + M_3^2 = \dfrac{3\left(a^2 + b^2 + c^2\right)}{4}$

Now apply AM > GM on M_1^2, M_2^2, M_3^2

$\Rightarrow \dfrac{\frac{3}{4}(a^2 + b^2 + c^2)}{3} > (M_1M_2M_3)^{2/3}$

since $a > b > c \Rightarrow M_1 < M_2 < M_3$

$\therefore M_1, M_2, M_3$ are in G.P.

$\Rightarrow M_1M_2M_3 = (M_2)^3$

$\therefore M_2 = \dfrac{1}{2}$ so

$\dfrac{a^2 + b^2 + c^2}{4} > \left(\dfrac{1}{2}\right)^{\frac{2}{3}\times 3} \Rightarrow a^2 + b^2 + c^2 > 1$

51. (5)

The given expression can be re written as

$\sum\limits_{x\in\{p,q,r,s,\}} \left(2x^2 + (1-x)^2\right)$ where $0 \le x \le 1$

consider the function $f(x) = 2x^2 + (1-x)^2$, $0 \le x \le 1$

So $f'(x) = 4x - 2(1 - x)$

For critical points

$f'(x) = 0$

or $x = 1/3$

Thus set of critical points is $\left\{0, \dfrac{1}{3}, 1\right\}$

Now $f(0) = 1$, $f\left(\dfrac{1}{3}\right) = \dfrac{2}{3}$, $f(1) = 2$

Thus maximum value of f(x) is 2 and minimum value of f(x) is 2/3. Therefore maximum value of given expression is $4 \times 2 = 8$ and minimum value is $4\times (2/3) = \dfrac{8}{3}$

52. (9)

Let x_1, x_2, x_3, x_4, x_5 are the no of objects obtained by different person

Then $x_1 + x_2 + x_3 + x_4 + x_5 = 21$

$1 \le x_1 < x_2 < x_3 < x_4 < x_5$

Let $y_1 = x_1$

$y_2 = x_2 - x_1$

$y_3 = x_3 - x_2$

$y_4 = x_4 - x_3$

$y_5 = x_5 - x_4$

$\therefore 5y_1 + 4y_2 + 3y_3 + 2y_4 + y_5 = 21$

$5t_5 + 4t_4 + 3t_3 + 2t_2 + t_1 = 6$ $\quad (t_i = y_i - 1, 2,3,4,5)$

$t_1, t_2, t_3, t_4, t_5 \ge 0$

the no. of solution of above equation is

Coeff. of x^6 in $(1 + x^5 + x^{10} + \ldots)(1 + x^4 + x^8 + \ldots.)(1 + x^3 + x^6 + \ldots)(1 + x^2 + x^4 + \ldots..)$ $(1 + x + x^2 + \ldots.)$

$=$ Coeff. of x^6 in $(1+x^5)(1+x^4)(1+x^3 + x^6)(1 + x^2 + x^4 + x^6)(1 + x + x^2 + \ldots..+ x^6)$

=Coeff , of x^6 in $(1+x^4 + x^5)$ $(1+ x^3 + x^6)$ $(1+ x^2 + x^4 + x^6)$ $(1+x+ x^2+ \ldots. +x^6)$
= Coeff. of x^6 in $(1+ x^3 + x^4 + x^5 + x^6)$ $(1+ x^2 + x^4 + x^6)$ $(1+x + x^2 + \ldots + x^6)$
= Coeff. pf x^6 in $(1+ x^2 + x^3 + 2x^4 + 2x^5+ 3x^6)$ $(1+x+ x^2+ \ldots + x^6)$
= 9
Now the variables $x_1, x_2, \ldots., x_5$ can permute in 5! ways
$\therefore$ Total no of ways $= 9\times 5!$
$= 1080$

53. (1)

Roots of the equation $ax^3+x-1-a = 0$ are $1,\alpha,\beta$

$$\therefore \alpha + \beta = -1 \text{ and } \alpha\beta = \frac{1+a}{a}$$

$$\Rightarrow \lim_{x\to\frac{1}{\alpha}} \frac{(1 + a)x^3 - x^2 - a}{(e^{1 - \alpha x} -1)(x - 1)} = \lim_{x\to\frac{1}{\alpha}} \frac{(x-1)[(1 + a) x^2 + ax + a]}{(e^{1 - \alpha x} -1)(x - 1)} = \lim_{x\to\frac{1}{\alpha}} \frac{[a\alpha\beta x^2 + ax + a]}{(e^{1 - \alpha x} -1)}$$

$$\Rightarrow \lim_{x\to\frac{1}{\alpha}} \frac{a(1 - \alpha x)(1 - \beta x)}{(e^{1 - \alpha x} -1)} = \lim_{x\to\frac{1}{\alpha}} a(1 - \beta x) = \frac{a(\alpha - \beta)}{\alpha}$$

54. (4)

α, β, γ are the roots of equation $x^3 + px^2 + qx - q = 0$ .....(i)

then equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ will be

$qx^3 - qx^2 - px - 1 = 0 \left(\text{after replacing } x \text{ by } \frac{1}{x}\right)$ .....(ii)

roots of this equation α', β', γ' are in A.P.

$2\beta' = \alpha' + \gamma'$

also $\alpha' + \beta' + \gamma' = 1$ $\qquad 3\beta' = 1$

$\alpha'\beta' + \beta'\gamma' + \alpha'\gamma' = -\frac{p}{q} \quad \Rightarrow \quad \beta' = \frac{1}{3}$

$\alpha'\beta'\gamma' = \frac{1}{q}$ $\qquad$ Put $x = \frac{1}{3}$ in equation (ii)

$9p + 2q = -27$

Using $\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)^2 = \frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}+2\left(\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\alpha\gamma}\right)$

Now $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=1$ also $\frac{1}{\beta}=\frac{1}{3}.$

So $1=\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}+2\left(\frac{1}{\beta}\left(\frac{1}{\alpha}+\frac{1}{\gamma}\right)+\frac{1}{\alpha\gamma}\right), \quad 1=\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}+2\left(\frac{2}{\beta^2}+\frac{1}{\alpha\gamma}\right)$

Now let $\frac{1}{\alpha}=\frac{1}{3}-d$, $\frac{1}{\gamma}=\frac{1}{3}+d$, $1=\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}+2\left(\frac{2}{9}+\left(\frac{1}{3}-d\right)\left(\frac{1}{3}+d\right)\right)$

$1=\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}+2\left(\frac{3}{9}-d^2\right), \quad 2d^2=\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}-\frac{1}{3}\geq 0, \quad$ so $\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\geq\frac{1}{3}$